Sensor and Systems
Sensor fusion

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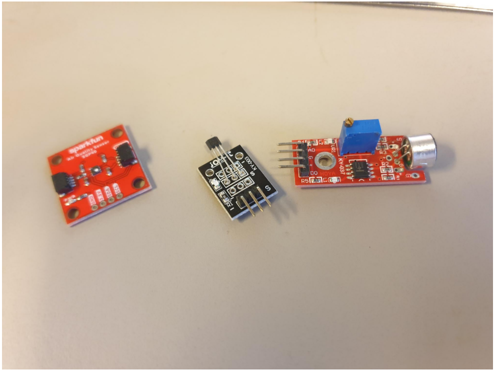
Electronic Systems

8th semester

Overview

- Sensor fusion
- Kalman filter for sensor fusion
- Complementary filter
- Attitude determination
  - Madgwick filter
  - Mahoney filter

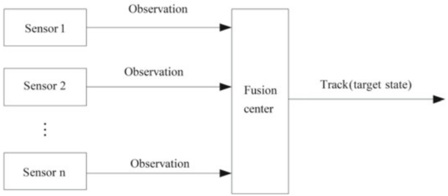
Simple sensors
... all the way back from the start of the course



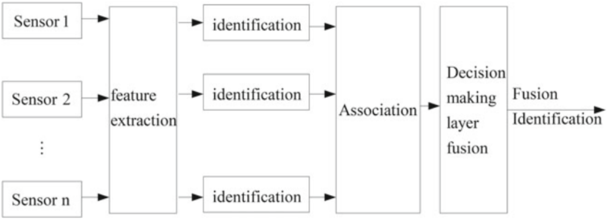
“Simple” sensors (From left: air quality, Hall effect, microphone)

- Measure one continuous physical quantity;
- Provide a simple analog or digital interface;
- Do not require extensive processing to obtain the desired information.

Sensor fusion



Sensor fusion is a broad term referring to using several sensors at the same time to provide information for a united purpose – e.g., estimating a common state.

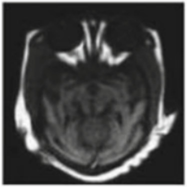


Decision making processes often make use of several different sensors to achieve more reliable feature detection – e.g., autonomous cars that rely on both radar, LiDAR, and cameras to detect obstacles, road signs etc.

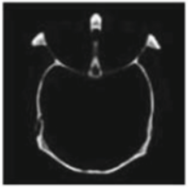
(a) focus left

(b) focus right

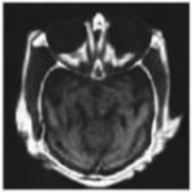
(c) fusion of (a) and (b)



(d) MRI

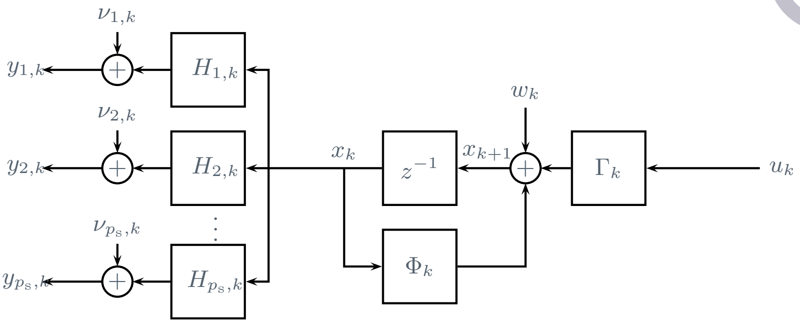


(e) CT



(f) fusion of (d) and (e)

Measurement noise affecting discrete-time system



- Discrete-time model with multiple sensor measurements:

$$
\begin{aligned}
x_{k+1} &= \Phi_k x_k + \Gamma_k u_k + w_k \\
y_{1,k} &= H_{1,k} x_k + \nu_{1,k} \\
y_{2,k} &= H_{2,k} x_k + \nu_{2,k} \\
&\vdots \\
y_{p_\mathrm{s},k} &= H_{p_\mathrm{s},k} x_k + \nu_{p_\mathrm{s},k}
\end{aligned}
$$

- Initial state and covariance estimates are $x_0 \in \mathcal{N}(\hat{x}_{0|-1}, P_{0|-1})$, as usual.
- The process noise is – also as usual – assumed to be Gaussian and independently distributed:

$$w_k \in \mathcal{N}(0, Q_k)$$

- The *measurement* noise is assumed to be white for each sensor, but not necessarily uncorrelated between individual sensors:

$$\nu_k \in \mathcal{N}(0, R_k), \quad R_k = \begin{bmatrix} R_{11,k} & R_{12,k} & \dots & R_{1p_\mathrm{s},k} \\ R_{12,k} & R_{22,k} & \dots & R_{2p_\mathrm{s},k} \\ \vdots & \vdots & \ddots & \vdots \\ R_{p_\mathrm{s}1,k} & R_{p_\mathrm{s}2,k} & \dots & R_{p_\mathrm{s}p_\mathrm{s},k} \end{bmatrix}.$$

- If we know R_k, it is often convenient to change measurement coordinates such that the cross-correlations are eliminated.
- Factorize R_k as
$$R_k = L_k^\top L_k$$
with L_k being lower triangular (Cholesky factorization).
- Let $T_k = (L_k^{-1})^\top$ and stack the output equations:
$$y_k = \begin{bmatrix} y_{1,k} \\ \vdots \\ y_{n_\mathrm{p},k} \end{bmatrix}, \quad H_k = \begin{bmatrix} H_{1,k} \\ \vdots \\ H_{n_\mathrm{p},k} \end{bmatrix}, \quad \nu_k = \begin{bmatrix} \nu_{1,k} \\ \vdots \\ \nu_{n_\mathrm{p},k} \end{bmatrix}.$$
- Define new outputs
$$\bar{y}_k = T_k y_k = T_k(H_k x_k + \nu_k) = \bar{H}_k x_k + \bar{\nu}_k$$
where
$$\bar{\nu}_k \in \mathcal{N}(0, \bar{R}_k), \quad \bar{R}_k = T_k R_k T_k^\top = (L_k^{-1})^\top L_k^\top L_k L_k^{-1} = I.$$

Kalman filter for sensor fusion, transformed measurements

- Initial conditions: $x_0 \in \mathcal{N}(\hat{x}_{0|-1}, P_{0|-1})$.
- Measurement update[1] after receiving y_k and u_k:

$$
\begin{aligned}
T_k &= \mathrm{chol}(R_k) \\
\hat{\bar{y}}_{k|k-1} &= \bar{H}_k \hat{x}_{k|k-1} \\
\tilde{y}_{k|k-1} &= T_k y_k - \hat{\bar{y}}_{k|k-1} \\
K_k &= P_{k|k-1} \bar{H}_k^\top (\bar{H}_k^\top P_{k|k-1} \bar{H}_k^\top + \bar{R}_k)^{-1} \\
\hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} \\
P_{k|k} &= (I - K_k \bar{H}_k)^{-1} P_{k|k-1} (I - K_k \bar{H}_k)^\top + K_k \bar{R}_k K_k^\top .
\end{aligned}
$$

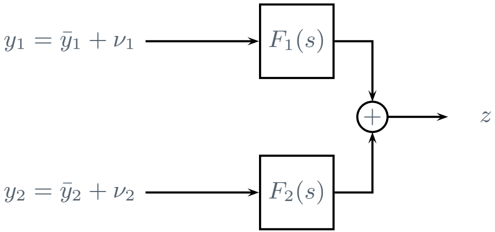
- Time update from k to $k+1$:

$$
\begin{aligned}
\hat{x}_{k+1|k} &= \Phi_k \hat{x}_{k|k} + \Gamma_k u_k \\
P_{k+1|k} &= \Phi_k P_k \Phi_k + Q_k
\end{aligned}
$$

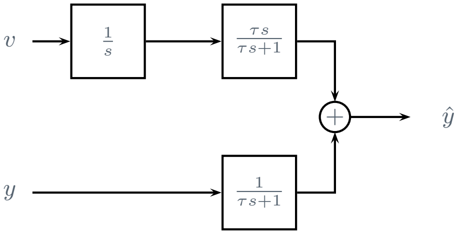
[1] Obs! Note that Yan et. al. use a different formulation for the $P_{k|k}$ update, namely

$$P_{k|k} = (I - K_k \bar{H}_k) P_{k|k-1}$$

The two expressions are, in fact, identical, although it takes a healthy dose of algebra to show it.



- $F_1(s), F_2(s)$: filters
- $z = F_1(s)y_1 + F_2(s)y_2$: sum of filtered versions of both measurements.
- Complementary filter: $F_1(s) = I - F_2(s)$.
- Allows separation of filtering in different frequency ranges.



- Suppose v and y are velocity and position measurements of some moving object. Ideally these measurements are related via

$$\bar{y} = \frac{1}{s}v.$$

- Let $F_2(s) = \frac{1}{\tau s+1}$ be a low-pass filter. Then $F_1(s)$ becomes the high-pass filter

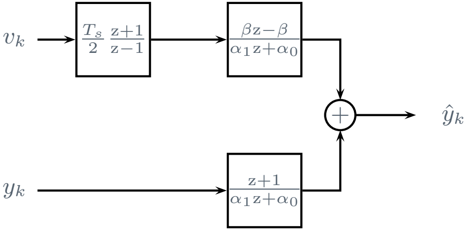
$$F_1(s) = 1 - \frac{1}{\tau s + 1} = \frac{\tau s}{\tau s + 1}.$$

- Position estimate:

$$z = \hat{y} = F_1(s)\frac{1}{s}v + F_2(s)y = \frac{1}{\tau s + 1}(\tau v + y).$$

Complementary filter

Discrete-time version



- Tustin approximation of integrator:

$$\bar{y}_k = \frac{T_s}{2}\frac{z+1}{z-1}v_k$$

- Low-pass filter: $F_2(z) = \frac{z+1}{\alpha_1 z+\alpha_0}$ where $\alpha_1 = \frac{\tau T_s}{2} + 1, \alpha_0 = 1 - \frac{\tau T_s}{2}$.
- High-pass complement: $F_1(z) = \frac{\beta(z-1)}{\alpha_1 z+\alpha_0}$ where $\beta = \alpha_1 - 1$.
- Position estimate:

$$\hat{y}_k = \frac{1}{\alpha_1}\left(-\alpha_0\hat{y}_{k-1} + \beta\frac{T_s}{2}v_k + \beta\frac{T_s}{2}v_{k-1} + y_k + y_{k+1}\right)$$

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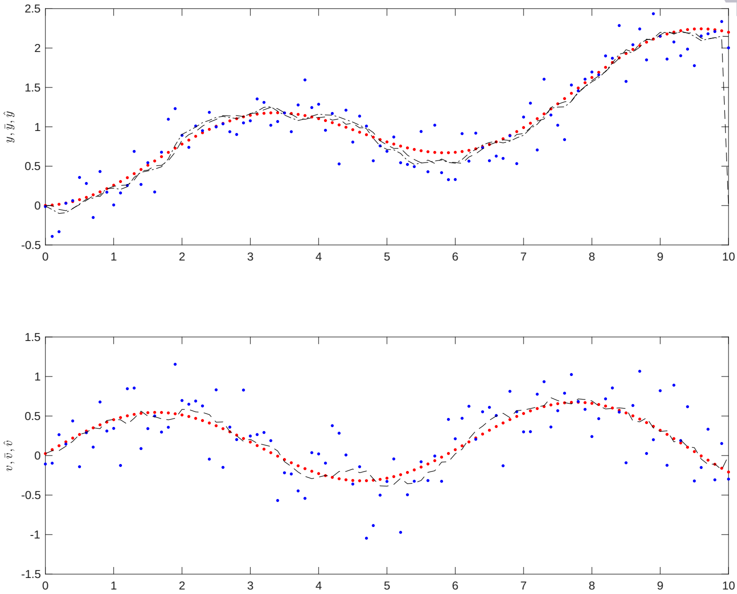
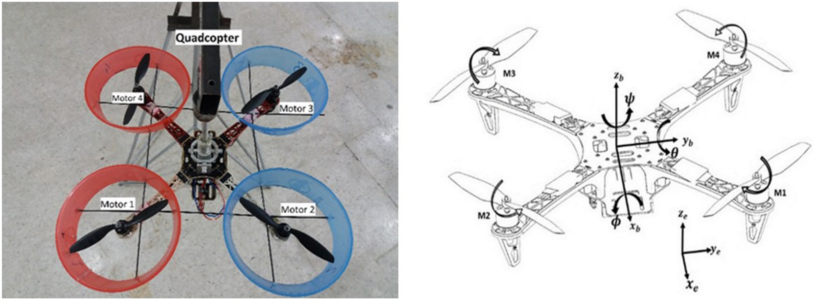
Filtering performance

Red: deterministic; blue: measurements, black: Complementary/Kalman filter est.



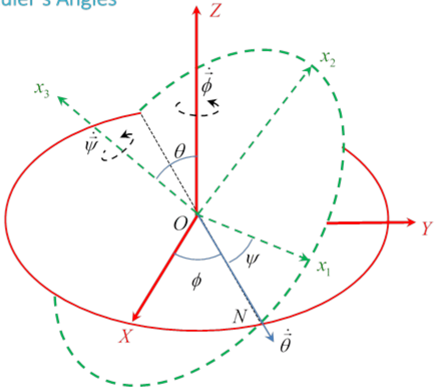
(a) (b)



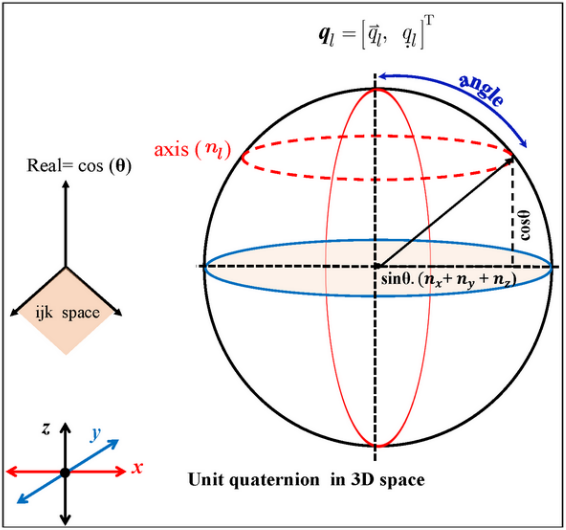
A common situation where sensor fusion gives clear benefits is *attitude determination*, for instance using IMUs.

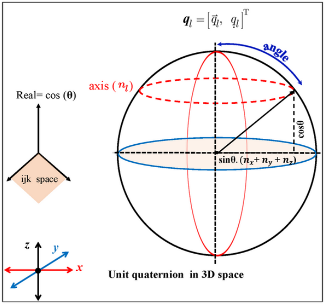
- Gyroscopes measure rate of rotation directly, typically in rad/s.
- Accelerometers measure the sum of gravity and acceleration in m/s^2.
- Magnetometers measure the magnetic field, typically in μT.

Euler's Angles



θ,ϕ follow standard physics practice for labeling the direction of body axis x_3 relative to lab axes X,Y,Z, ψ is the body rotation angle from ON to the x_1 axis in the x_1,x_2 plane, about its x_3 axis.


$q_l = [\vec{q}_l,\ q_l]^T$
angle
axis (n_l)
Real= cos (θ)
cosθ
sinθ. (n_x+ n_y + n_z)
ijk space
z
y
x
Unit quaternion in 3D space



- Quaternions are extensions to complex numbers with a '3D imaginary part':

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \vec{q} \end{bmatrix} \in \mathbb{R}^4$$

- A unit quaternion has magnitude 1:

$$\|\mathbf{q}\| = \sqrt{q_0^2 + q_x^2 + q_y^2 + q_z^2} = 1$$

- Unit quaternions can represent rotations effectively.

Assuming a quaternion is properly normalized, the corresponding Euler angles can be obtained via the relations:

$$\begin{bmatrix}\phi\\ \theta\\ \psi\end{bmatrix} = \begin{bmatrix} \operatorname{atan2}\left(2(q_0q_x + q_yq_z), 1 - 2(q_x^2 + q_y^2)\right) \\ -\pi/2 + 2\operatorname{atan2}\left(\sqrt{1 + 2(q_0q_y - q_xq_z)}, \sqrt{1 - 2(q_0q_y - q_xq_z)}\right) \\ \operatorname{atan2}\left(2(q_0q_z + q_xq_y), 1 - 2(q_y^2 + q_z^2)\right)\end{bmatrix}$$

Rotation via multiplication by unit quaternion

- Multiplication of two quaternions $\mathbf{q} = (q_0, \vec{q})$ and $\mathbf{r} = (r_0, \vec{r})$ is given by the formula

$$\mathbf{q} \otimes \mathbf{r} = \begin{bmatrix} q_0 r_0 - \vec{q} \cdot \vec{r} \\ q_0 \vec{r} - r_0 \vec{q} + \vec{q} \times \vec{r} \end{bmatrix}$$

  where $\cdot$ and $\times$ indicate the usual three-dimensional dot and vector cross products.
- The canonical way to rotate a three-dimensional vector $\vec{v}$ by a quaternion $\mathbf{q}$ defining an Euler rotation is via the formula

$$\mathbf{v}' = \mathbf{q} \otimes \mathbf{v} \otimes \mathbf{q}^*$$

  where $\mathbf{v} = (0, \vec{v})$ is a quaternion containing the embedded vector $\vec{v}$, $\mathbf{q}^* = (q_0, -\vec{q})$ is the *conjugate* of $\mathbf{q}$, and $\mathbf{v}' = (0, \vec{v}')$ gives the rotated vector $\vec{v}'$.
- An alternative approach is to apply the equations

$$\begin{aligned} \vec{t} &= 2\vec{q} \times \vec{v} \\ \vec{v}' &= \vec{v} + q_0 \vec{t} + \vec{q} \times \vec{t}. \end{aligned}$$

- Note that

$$\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{\|\mathbf{q}\|}.$$

- Gyroscope measurement model[2]:

$$\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix} = \begin{bmatrix}\bar{\omega}_x\\ \bar{\omega}_y\\ \bar{\omega}_z\end{bmatrix} + \begin{bmatrix}\omega_{x,b}\\ \omega_{y,b}\\ \omega_{z,b}\end{bmatrix} + \begin{bmatrix}\nu_{\omega,x}\\ \nu_{\omega,y}\\ \nu_{\omega,z}\end{bmatrix}$$

  where $(\cdot)_b$ denotes bias and $\bar{\cdot}$ denotes 'true' angular rotation speed;
- Accelerometer measurement model:

$$\begin{bmatrix}a_x\\ a_y\\ a_z\end{bmatrix} = \begin{bmatrix}\bar{a}_x\\ \bar{a}_y\\ \bar{a}_z\end{bmatrix} + \begin{bmatrix}a_{x,b}\\ a_{y,b}\\ a_{z,b}\end{bmatrix} + \begin{bmatrix}\nu_{a,x}\\ \nu_{a,y}\\ \nu_{a,z}\end{bmatrix}$$

  where $\bar{a}$ is the sum of "gravity" and external acceleration;
- Magnetometer measurement model:

$$\begin{bmatrix}s_x\\ s_y\\ s_z\end{bmatrix} = \begin{bmatrix}\bar{s}_x\\ \bar{s}_y\\ \bar{s}_z\end{bmatrix} + \begin{bmatrix}s_{x,b}\\ s_{y,b}\\ s_{z,b}\end{bmatrix} + \begin{bmatrix}\nu_{s,x}\\ \nu_{s,y}\\ \nu_{s,z}\end{bmatrix}$$

[2] Note that the notation is changed from the one used in Ludwig and Burnham's paper

Model in quaternion formulation

- Kinematic equation:

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \boldsymbol{\omega}$$

where $\mathbf{q}$ represents orientation and $\boldsymbol{\omega} = (0, \vec{\omega})$ represents angular velocity.

- Acceleration quaternion representation rotated by $\mathbf{q}$:

$$\mathbf{a} = \mathbf{q}^{-1} \otimes \begin{bmatrix} 0 \\ \vec{a} \end{bmatrix} \otimes \mathbf{q}$$

(Note that in steady state, e.g. in hover, $\vec{a} = \begin{bmatrix} 0 & 0 & -g \end{bmatrix}^\top$)

- Magnetic field quaternion representation rotated by $\mathbf{q}$:

$$\mathbf{s} = \mathbf{q}^{-1} \otimes \begin{bmatrix} 0 \\ \vec{s} \end{bmatrix} \otimes \mathbf{q}.$$

Madgwick filter

- Measurements $\vec{\omega}_k$, $\vec{a}_k$ and $\vec{s}_k$ have been acquired; measurement update:

$$\hat{\mathbf{a}}_k = \hat{\mathbf{q}}_{k-1}^{-1} \otimes \mathbf{a} \otimes \hat{\mathbf{q}}_{k-1}$$
$$\hat{\mathbf{s}}_k = \hat{\mathbf{q}}_{k-1}^{-1} \otimes \mathbf{s} \otimes \hat{\mathbf{q}}_{k-1}$$

where $\mathbf{a}$ and $\mathbf{s}$ are reference quaternions.

- Compute correction terms:

$$\tilde{\vec{\omega}}_k = \vec{a}_k \times \hat{\vec{a}}_k + \vec{s}_k \times \hat{\vec{s}}_k$$
$$\hat{\vec{\omega}}_{b,k} = \hat{\vec{\omega}}_{b,k-1} - k_i T_s \tilde{\vec{\omega}}_k$$

- Estimation update:

$$\hat{\boldsymbol{\omega}}_k = \begin{bmatrix} 0 \\ \vec{\omega}_k \end{bmatrix} - \begin{bmatrix} 0 \\ \hat{\omega}_{b,k} \end{bmatrix} + \begin{bmatrix} 0 \\ k_p \tilde{\vec{\omega}}_k \end{bmatrix}$$
$$\hat{\mathbf{q}}_k = \hat{\mathbf{q}}_{k-1} + \frac{T_s}{2} \hat{\mathbf{q}}_{k-1} \otimes \hat{\boldsymbol{\omega}}_k .$$

- Measurements $\vec{\omega}_k$, $\vec{a}_k$ and $\vec{s}_k$ have been acquired:

$$\mathbf{h}_k = \hat{\mathbf{q}}_{k-1} \otimes \hat{\mathbf{s}}_k \otimes \hat{\mathbf{q}}_{k-1}^{-1}$$
$$\mathbf{m}_k = \begin{bmatrix} 0 & 0 & \sqrt{\hat{h}_{x,k}^2 + \hat{h}_{y,k}^2} & \hat{h}_{z,k} \end{bmatrix}^\top$$

- Compute error terms and gradient:

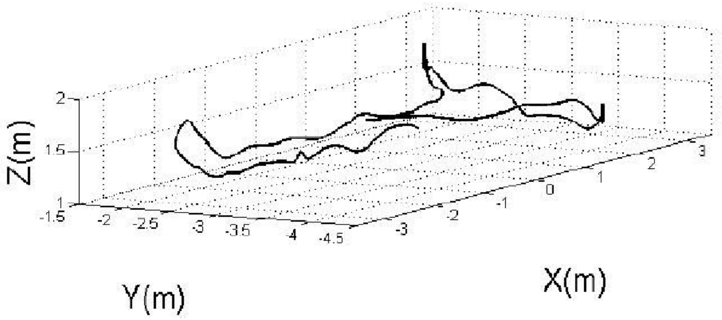
$$F_k = \begin{bmatrix} \hat{\mathbf{q}}_{k-1}^{-1} \otimes \mathbf{a} \otimes \hat{\mathbf{q}}_{k-1} - \mathbf{a}_k \\ \hat{\mathbf{q}}_{k-1}^{-1} \otimes \mathbf{s} \otimes \hat{\mathbf{q}}_{k-1} - \mathbf{s}_k \end{bmatrix}$$
$$\hat{\mathbf{q}}_{e,k} = J_k^\top F_k$$

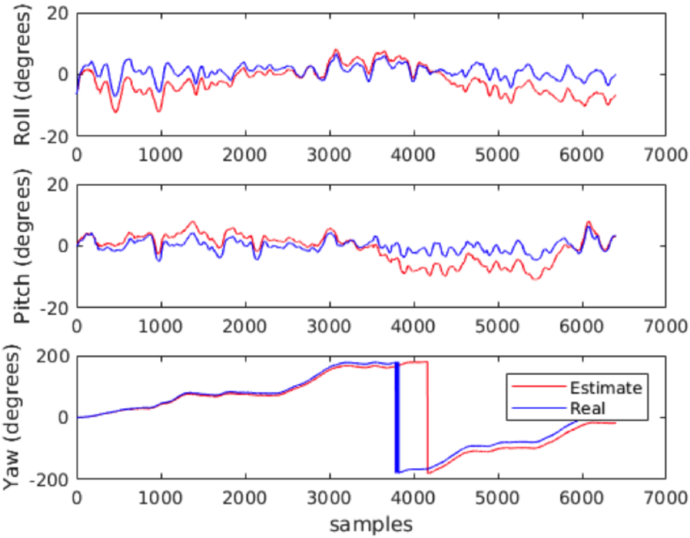
  where J_k is the Jacobian of F_k.

- Estimate update:

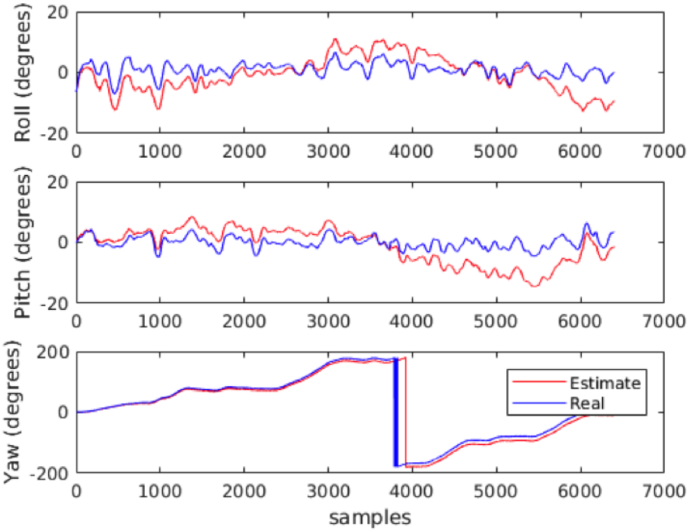
$$\hat{\boldsymbol{\omega}}_{e,k} = 2\hat{\mathbf{q}}_{k-1} \otimes \hat{\mathbf{q}}_{e,k}$$
$$\hat{\vec{\omega}}_{b,k} = \hat{\vec{\omega}}_{b,k-1} + T_s \hat{\omega}_{e,k}$$
$$\hat{\vec{\omega}}_k = \vec{\omega}_k - \zeta \hat{\vec{\omega}}_{b,k}$$
$$\hat{\mathbf{q}}_k = \hat{\mathbf{q}}_{k-1} + \frac{T_s}{2} \hat{\mathbf{q}}_{k-1} \otimes \hat{\boldsymbol{\omega}}_k - T_s \beta \frac{\hat{\mathbf{q}}_{e,k}}{\|\hat{\mathbf{q}}_{e,k}\|}.$$

Flight experiment





Roll (degrees)
20
0
-20
Pitch (degrees)
20
0
-20
Yaw (degrees)
200
0
-200
0
1000
2000
3000
4000
5000
6000
7000
Estimate
Real
samples


Roll (degrees)
Pitch (degrees)
Yaw (degrees)
samples
Estimate
Real
20
0
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4000
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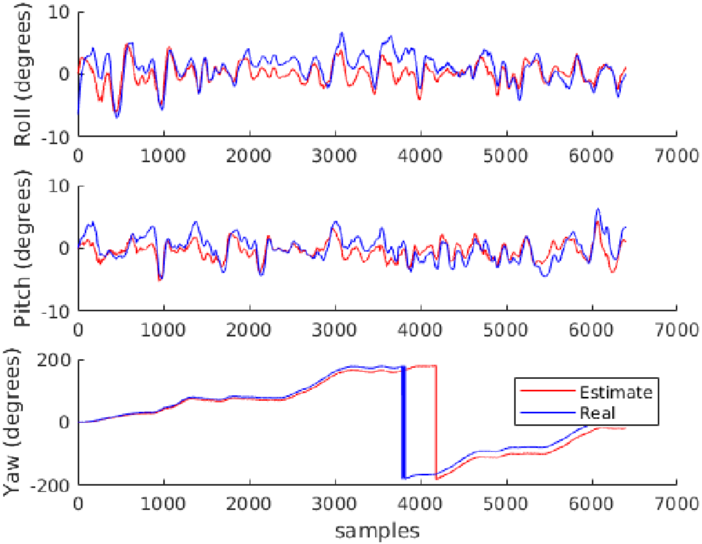
6000
7000

Filtering performance
Extended Kalman filter



- Sensor fusion is the concept of using more than one sensor to provide information about the same state.
- The Kalman filter handles this in a straightforward manner, but beware that the different sensors can give measurements in vastly different units.
- Correlation between measurement noise channels can be canceled by transforming the output coordinates appropriately
- Complementary filters are simple filters for sensor fusion of signals that are connected by integration/differentiation.
- Madgwick and Mahony filters are specifically designed for sensor fusion of IMU data; they make use of quaternions.